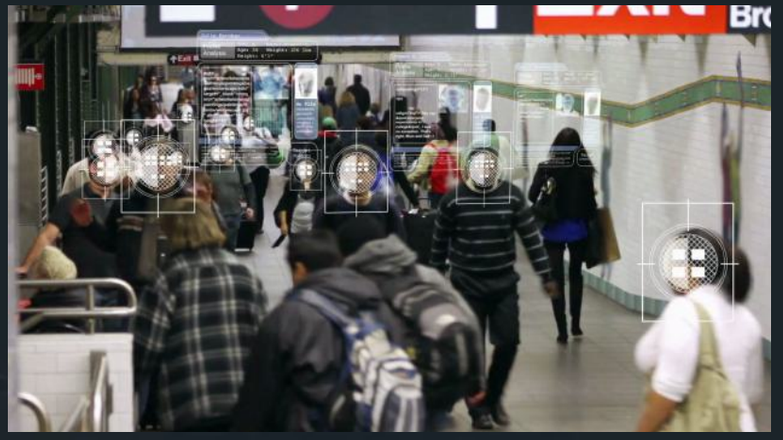
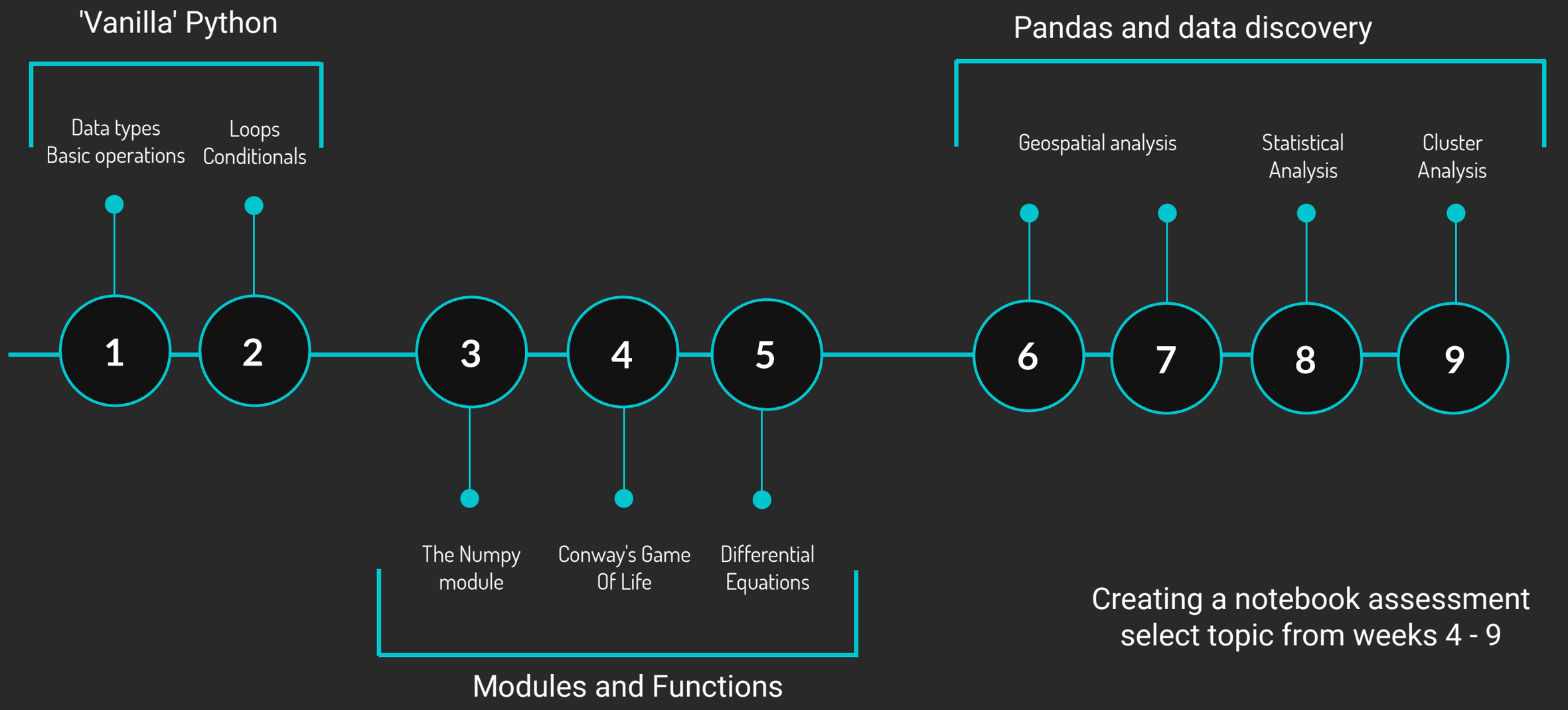
PANDAS AND DATA DISCOVERY



- MACHINE LEARNING AND CLUSTER ANALYSIS
- PRACTICAL 8



David Topping: david.topping@manchester.ac.uk

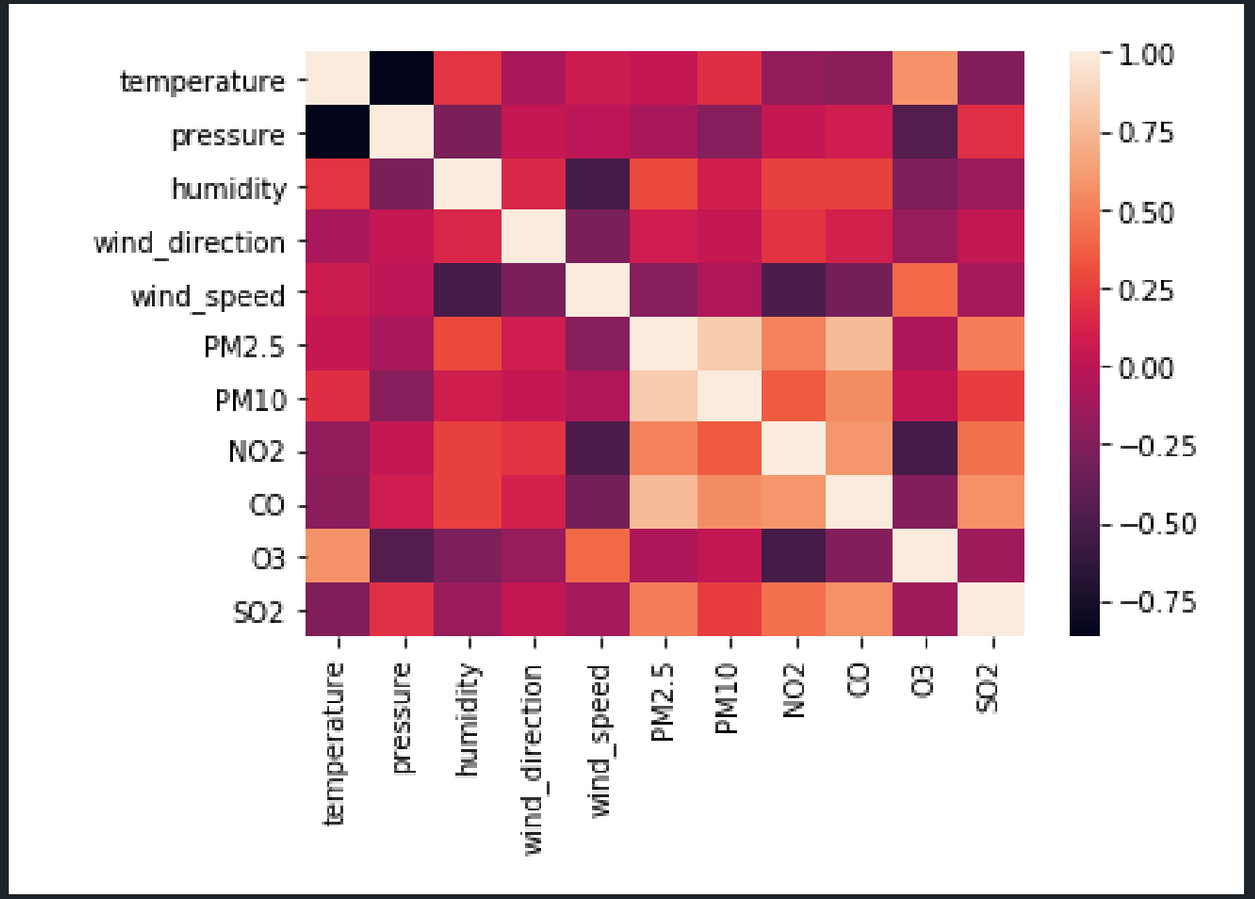


COURSE LAYOUT >>>>

LECTURE + PRACTICAL. SOLUTIONS ARE GIVEN

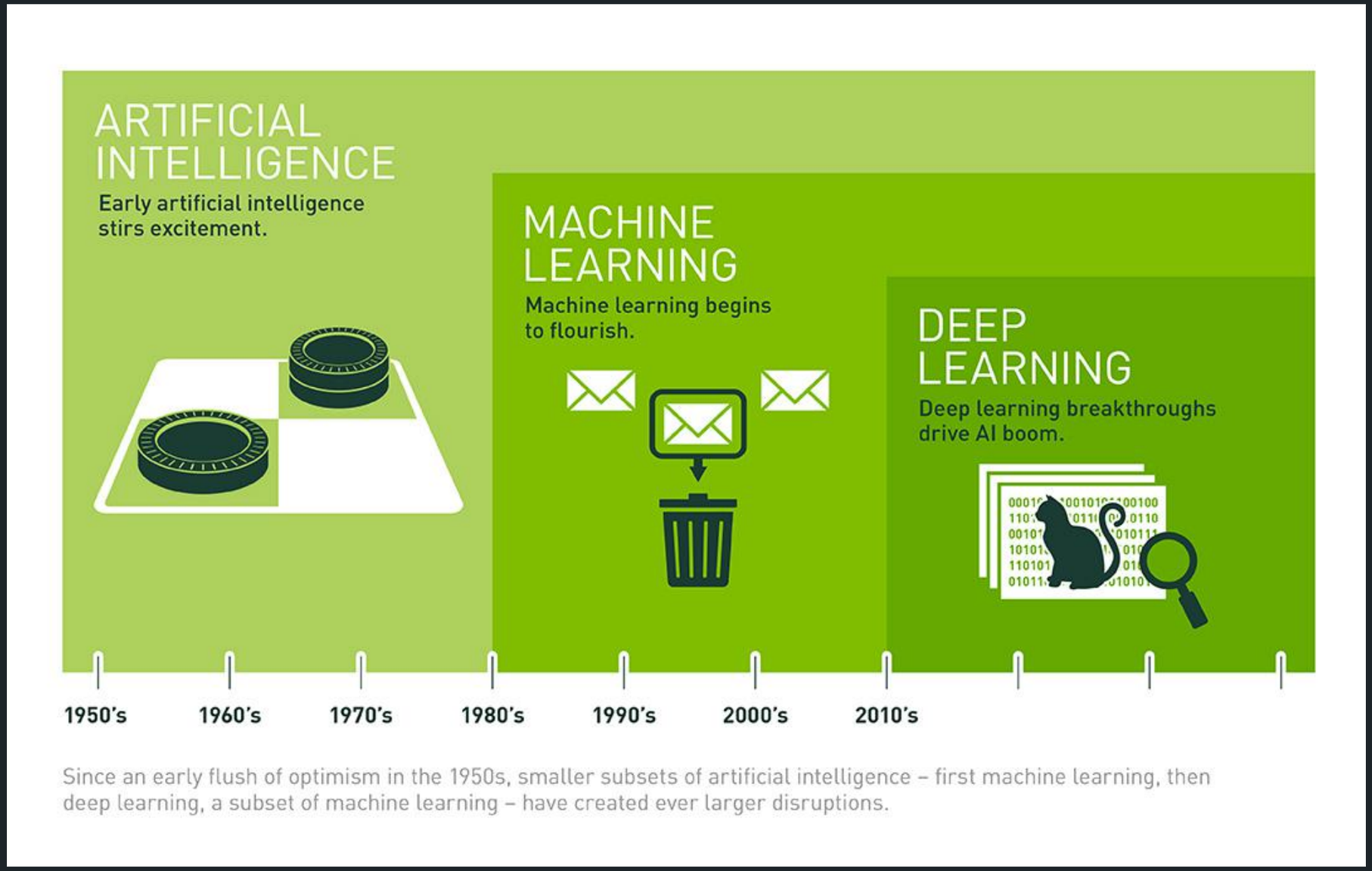
RECAP

- Pandas allows us to store information in a dataframe
- Pandas can deal with various data types
- We can perform statistical analysis on entire columns or subsets
- Dataframes have internal functions we can use to our advantage



What if we want to group similar observations together?...start using machine learning libraries

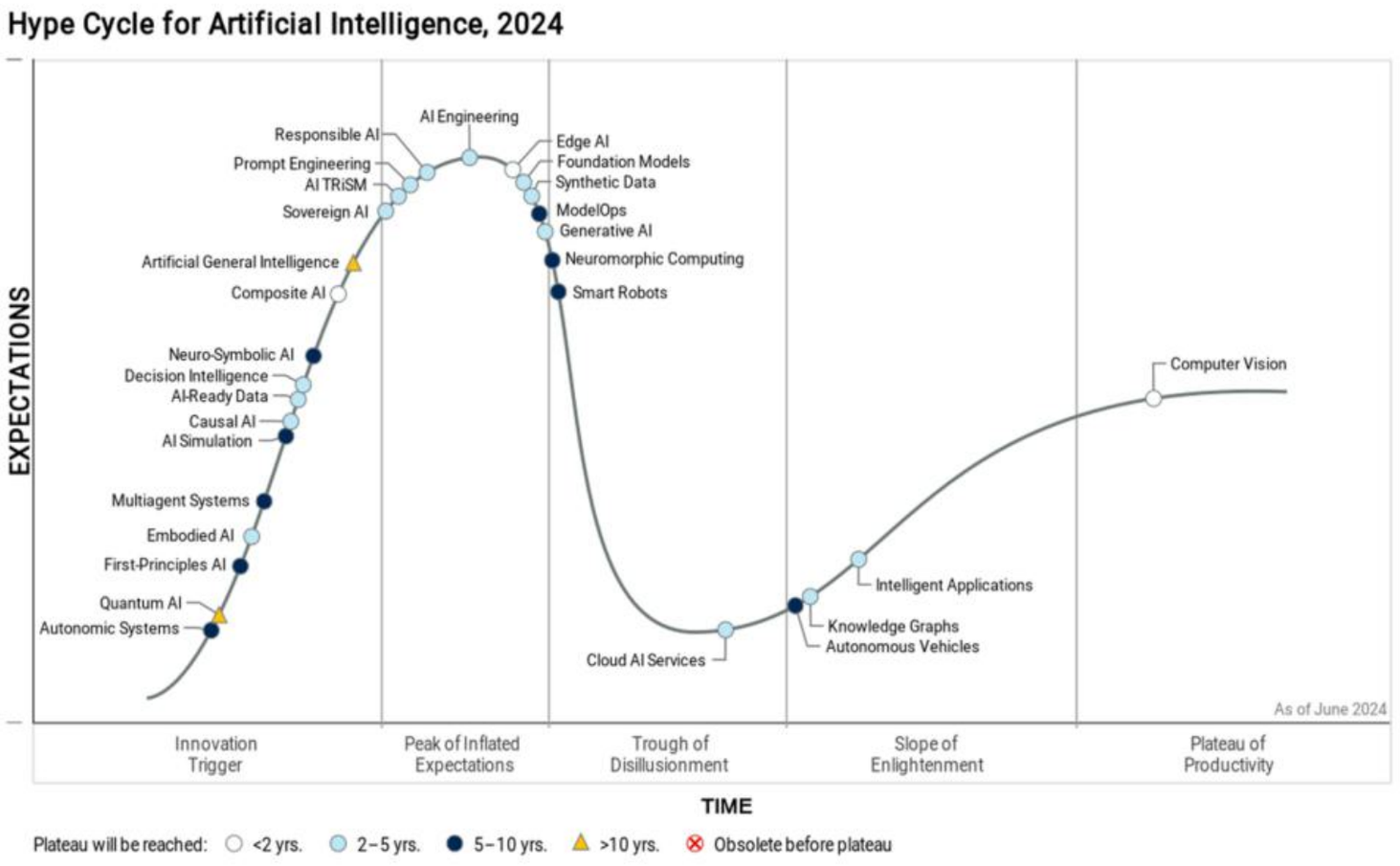
MACHINE LEARNING HAS A RICH HISTORY



https://news.developer.nvidia.com/on-demand-webinar-deep-learning-demystified/

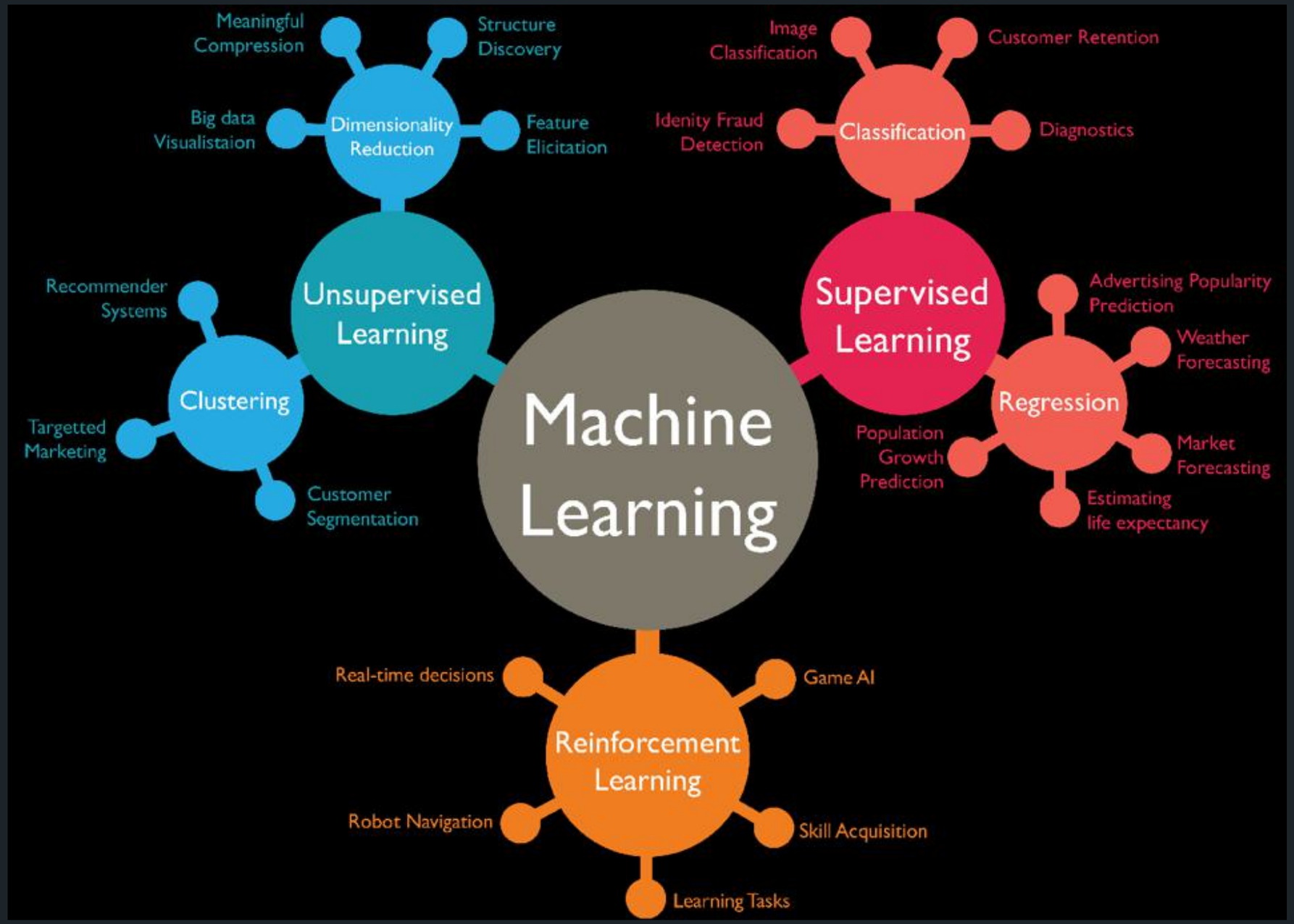
CONSTANTLY AT THE PEAK OF EXPECTATIONS??

Figure 1: Hype Cycle for Artificial Intelligence, 2024

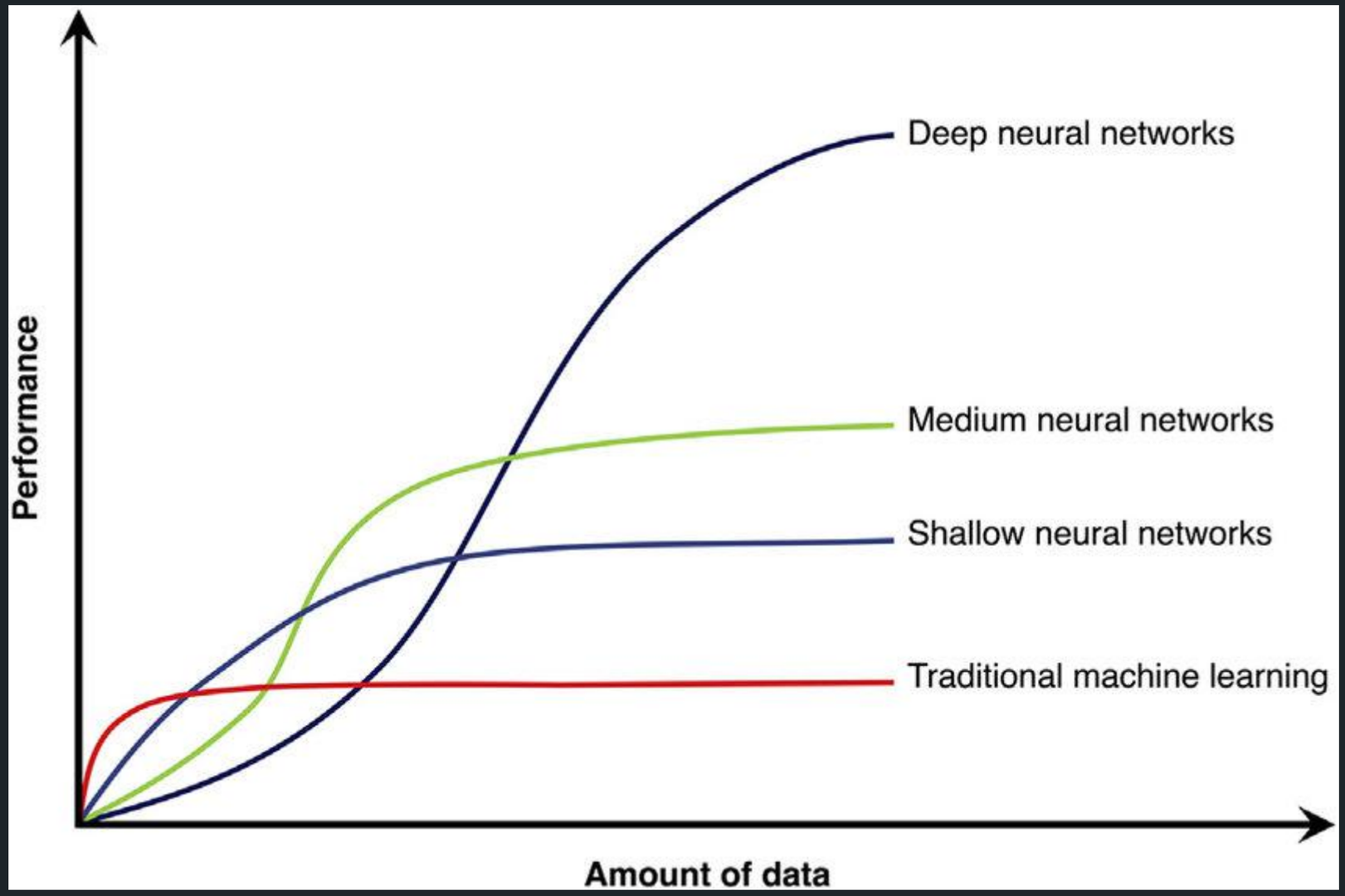


Gartner

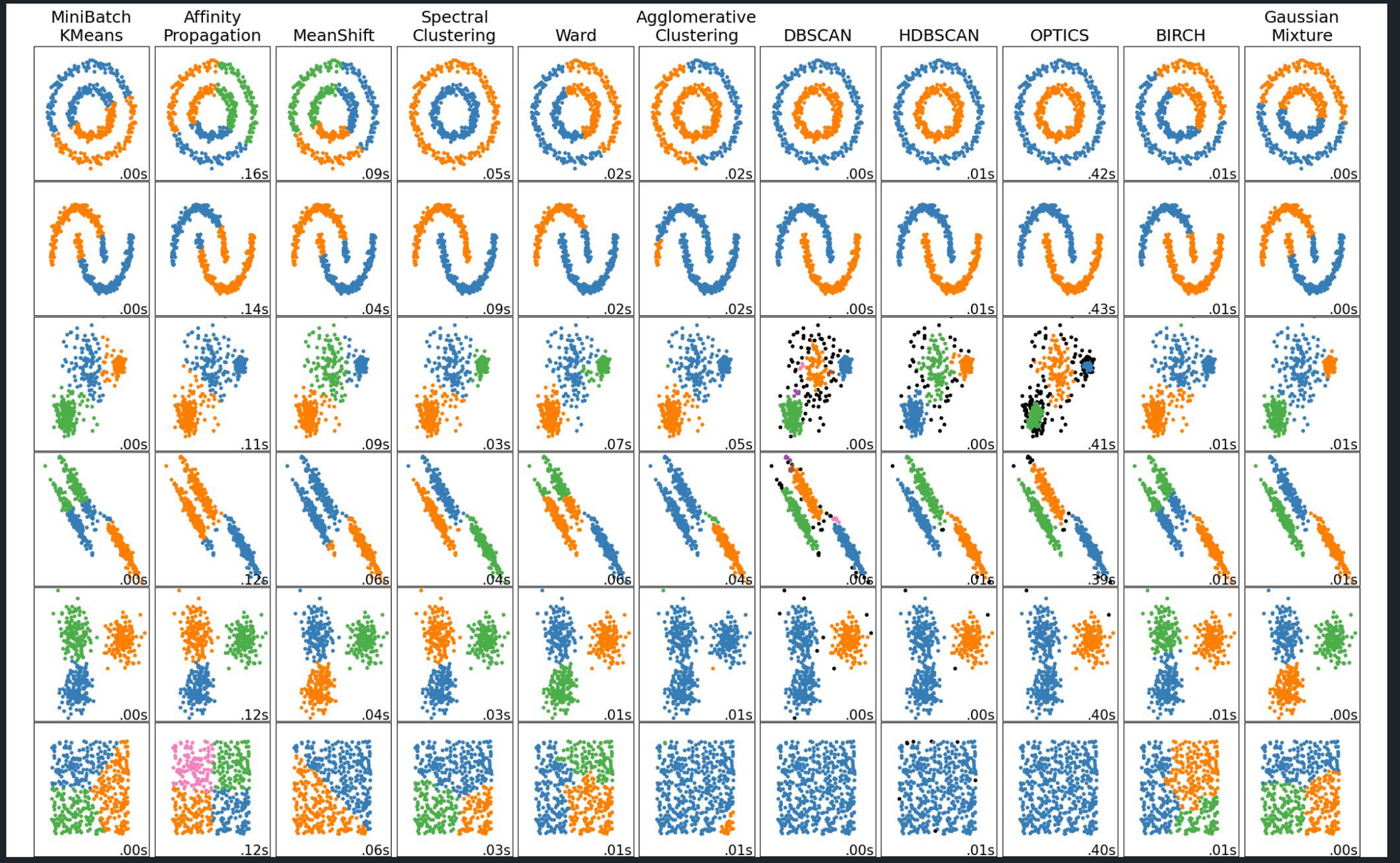
TYPES OF MACHINE LEARNING



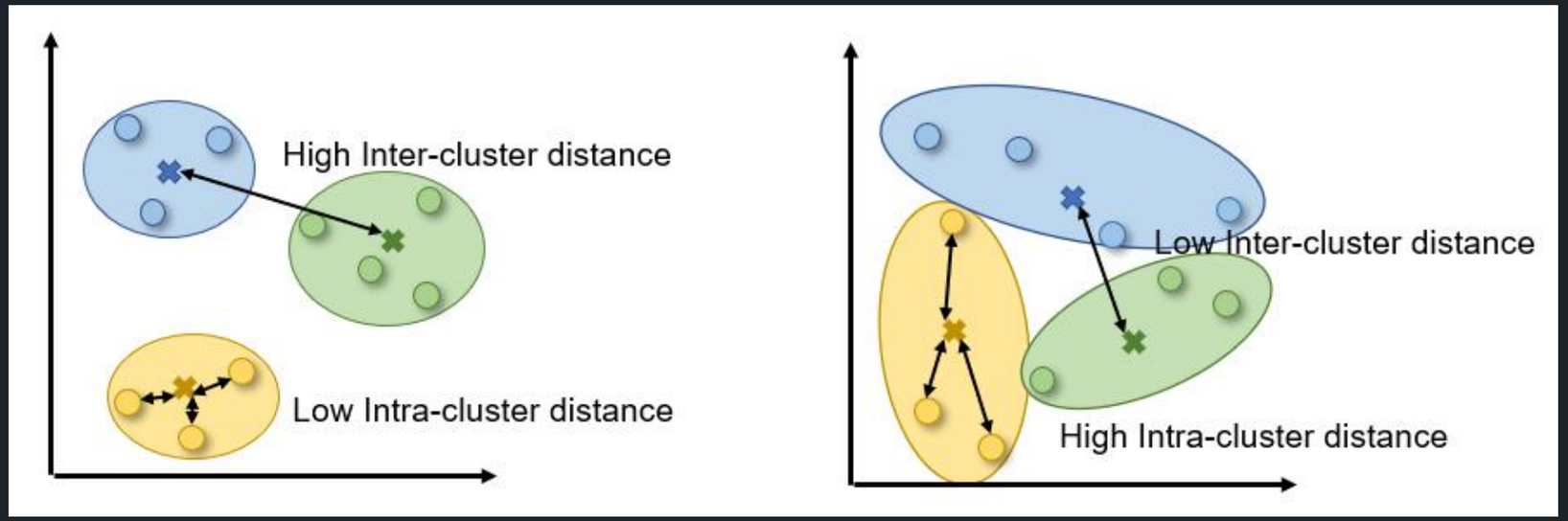
TYPES OF MACHINE LEARNING - DATA DEPENDENCY



CLUSTER ANALYSIS - UNSUPERVISED LEARNING



CLUSTER ANALYSIS - UNSUPERVISED LEARNING



'DISTANCES'...OR 'SIMILARITY' CAN BE CALCULATED IN DIFFERENT WAYS

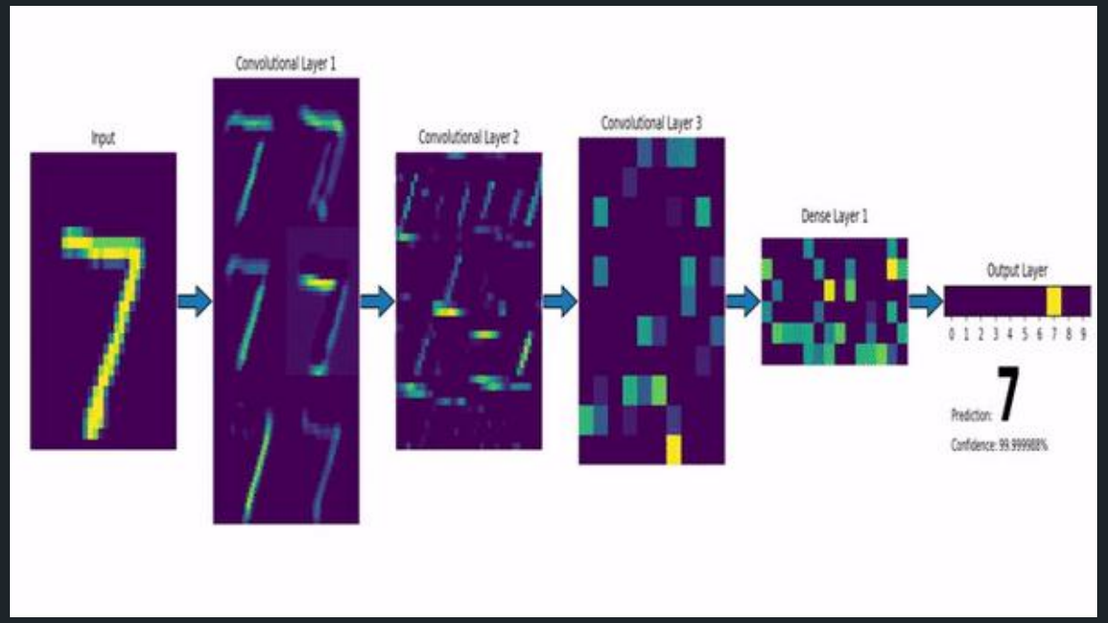
MACHINE LEARNING CONVERTS FEATURES INTO NUMBERS...

Euclidean $\sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$

Manhattan $\sum_{i=1}^{k}|x_i - y_i|$

Minkowski $\left(\sum_{i=1}^{k}(|x_i - y_i|)^q\right)^{1/q}$

NEURAL NETWORKS - SUPERVISED LEARNING



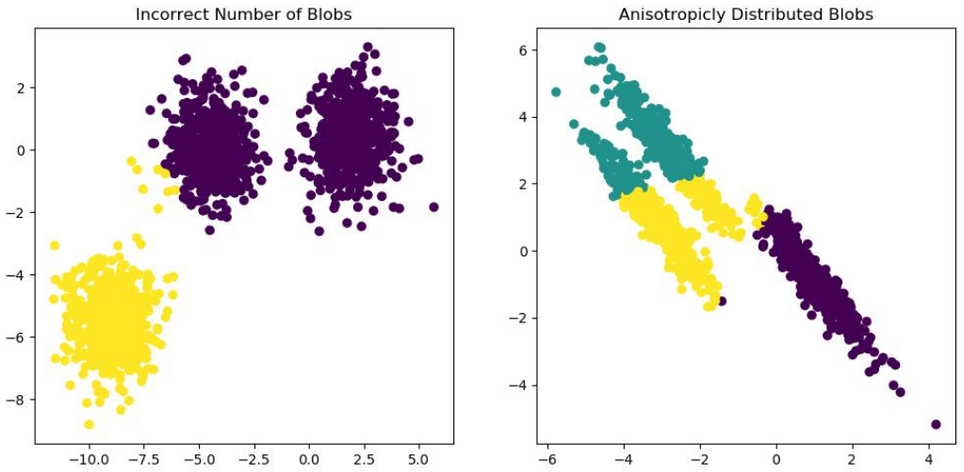
0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

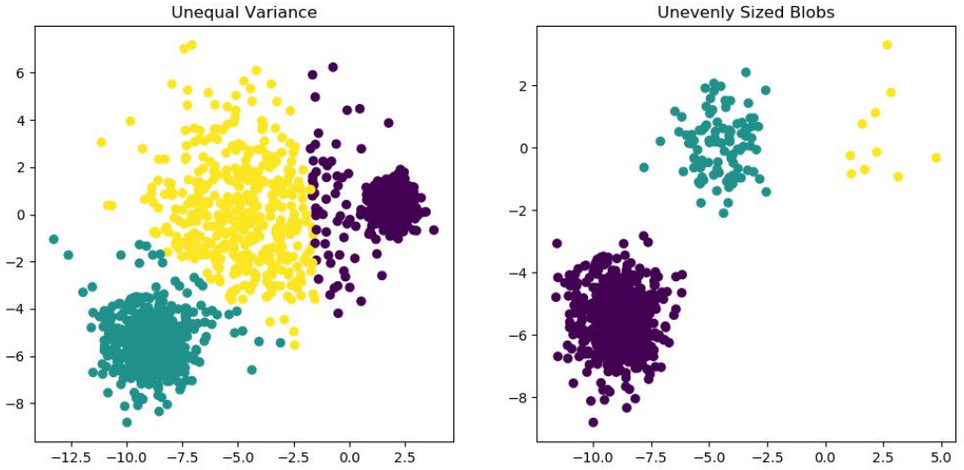
Kernel

0	-1	0
-1	5	-1
0	-1	0

114				

CLUSTER ANALYSIS - K MEANS

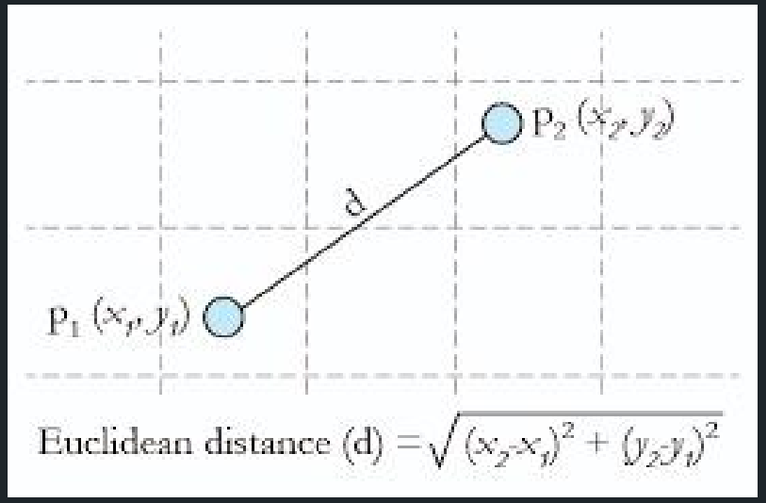




SOMETIMES WE CAN SEE DISTINCT GROUPINGS
IF WE HAVE MANY DIMENSIONS, THIS IS NOT SO EASY

CAN AN ALGORITHM USE NUMERICAL VALUES TO GROUP 'SIMILAR' OBSERVATIONS?

WHAT IF A DISTANCE COULD BE USED?



THERE IS A MODULE WE CAN USE.....WHAT A SURPRISE!!



In [1]:
```
import pandas as pd #Im using pd here as its easier to keep writing! You can use whatever you want, but it might help
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
import seaborn as sns
# Read data from file 'GVP_Eruption_Results.csv'
# We are going to use the function 'read_csv' within the Pandas package:

# Notice how we call that function using the '.' operator?
# (Note the data file needs to be in the same directory that your jupyter notebook is based) You can control delimite
data = pd.read_csv("data/forestfires.csv")

# How do we preview the data file.
# Preview the first 5 lines of the loaded data
data.head()
#data.columns.values
```

Out[1]:

	X	Y	month	day	FFMC	DMC	DC	ISI	temp	RH	wind	rain	area
0	7	5	mar	fri	86.2	26.2	94.3	5.1	8.2	51	6.7	0.0	0.0
1	7	4	oct	tue	90.6	35.4	669.1	6.7	18.0	33	0.9	0.0	0.0
2	7	4	oct	sat	90.6	43.7	686.9	6.7	14.6	33	1.3	0.0	0.0
3	8	6	mar	fri	91.7	33.3	77.5	9.0	8.3	97	4.0	0.2	0.0
4	8	6	mar	sun	89.3	51.3	102.2	9.6	11.4	99	1.8	0.0	0.0

EXTRACT DATA FROM A DATAFRAME TO PERFORM K-MEANS ON IT

- Extract our variables of interest from the dataframe into a new Numpy matrix
- Specify how many clusters we want the Kmeans algorithm to find
- Fit the clustering algorithm to our Numpy matrix
- Extract the labels to which each row in our matrix has been assigned.

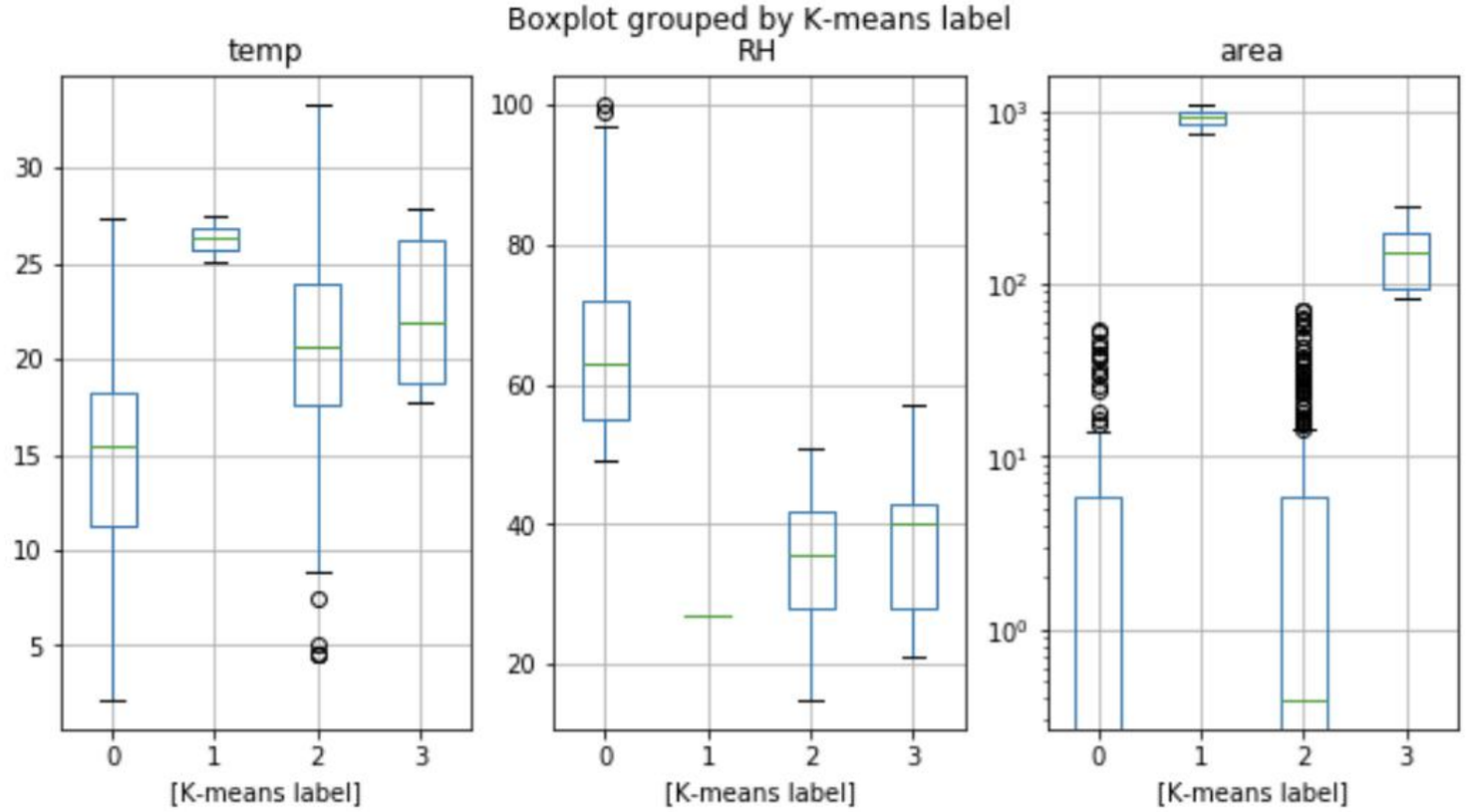
```
from sklearn.cluster import KMeans
# Extract our variables of interest from the dataframe into a new Numpy matrix
numpy_matrix = data[['temp','RH','area']].values
# Specify how many clusters we want the Kmeans algorithm to find
clusterer=KMeans(n_clusters=4)
# Fit the clustering algorithm to our Numpy matrix
clusterer.fit(numpy_matrix)
# Extract the labels to which each row in our matrix has been assigned.
labels = clusterer.labels_

# In the dataframe 'data' we can store the labels from using K-means:
data['K-means label'] = labels
# For example the following simply prints the new dataframe column to the screen
data['K-means label']
```

```
0      0
1      2
2      2
3      0
4      0
      ..
512    2
513    0
514    0
515    2
516    2
Name: K-means label, Length: 517, dtype: int32
```

VISUALISE THE PROPERTIES OF EACH DISTINCT GROUP

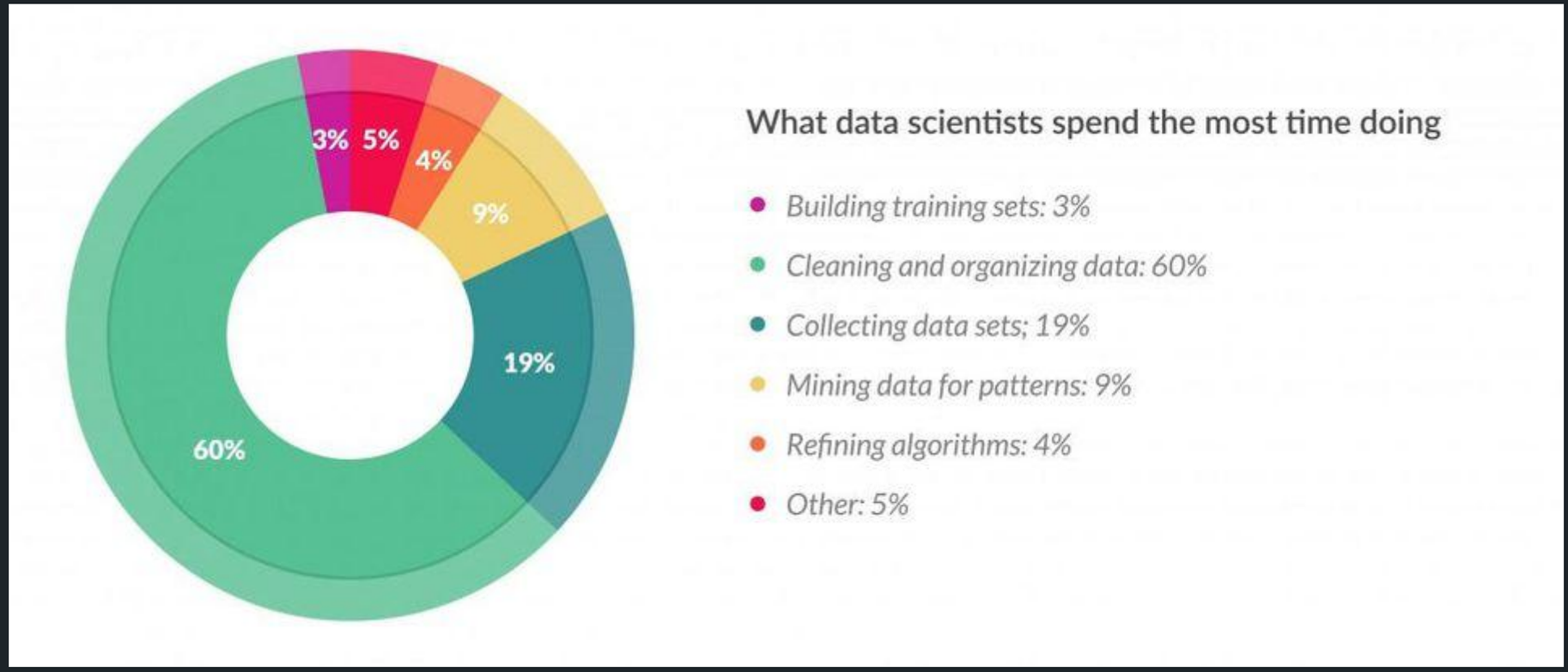
```
fig, ax = plt.subplots(1, 3, figsize=(10, 5))
data.boxplot(column=['temp'], by=['K-means label'], ax=ax[0])
data.boxplot(column=['RH'], by=['K-means label'], ax=ax[1])
data.boxplot(column=['area'], by=['K-means label'], ax=ax[2]).set_yscale('log')
```



HOWEVER, WE NEED TO UNDERSTAND OUR DATA BEFORE WE TRY AND CLUSTER IT.

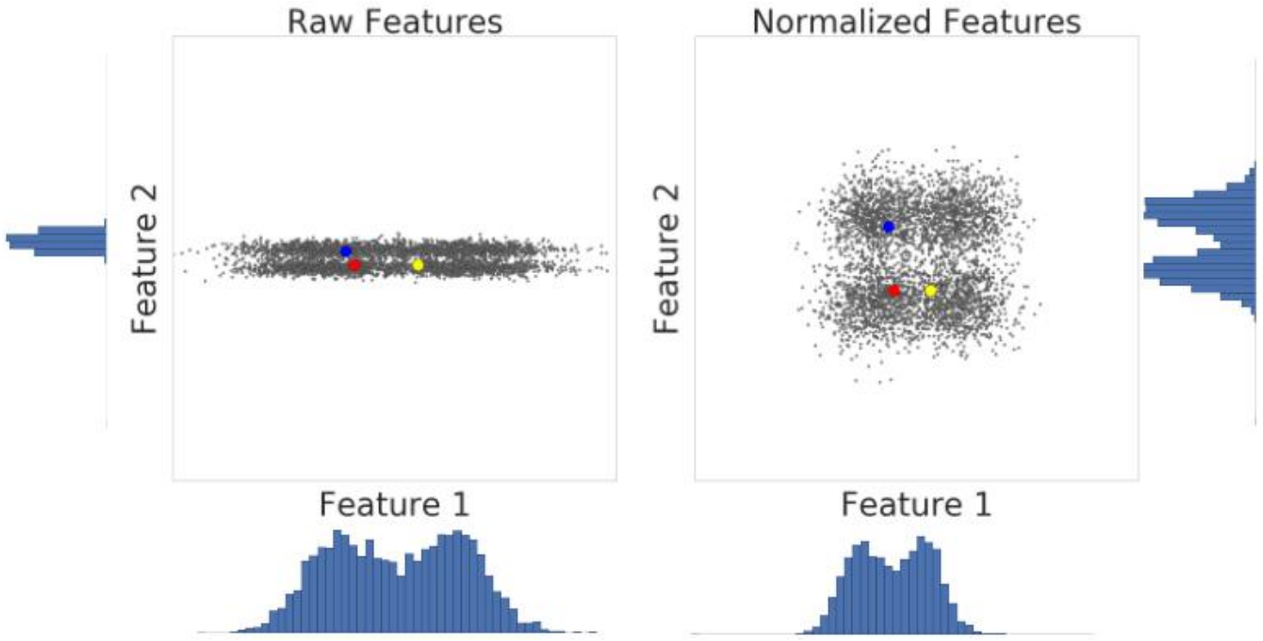
THERE IS A COMMON, OFTEN 'PAINFUL', PROCESS OF DATA PREPARATION BEFORE WE USE MACHINE LEARNING

DATA PREPARATION TAKES UP >90% OF THE MACHINE LEARNING PIPELINE

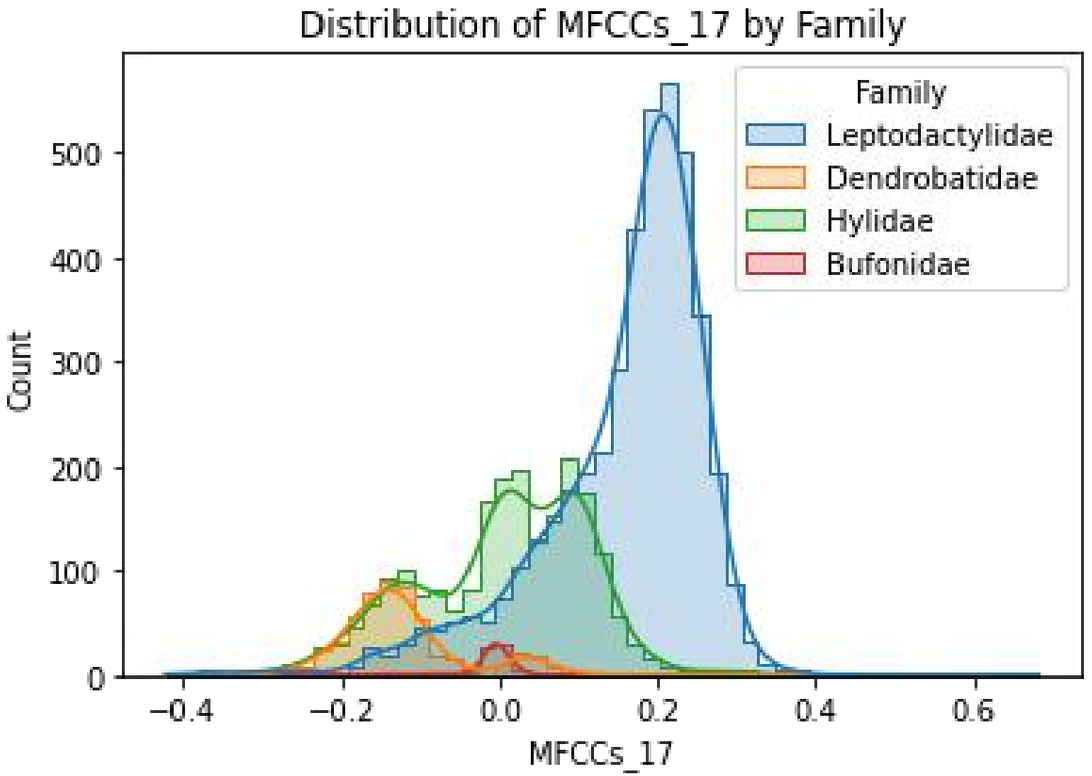


HOWEVER, WE NEED TO UNDERSTAND OUR DATA BEFORE WE TRY AND CLUSTER IT.

Distance based calculations can be dominated by raw feature range

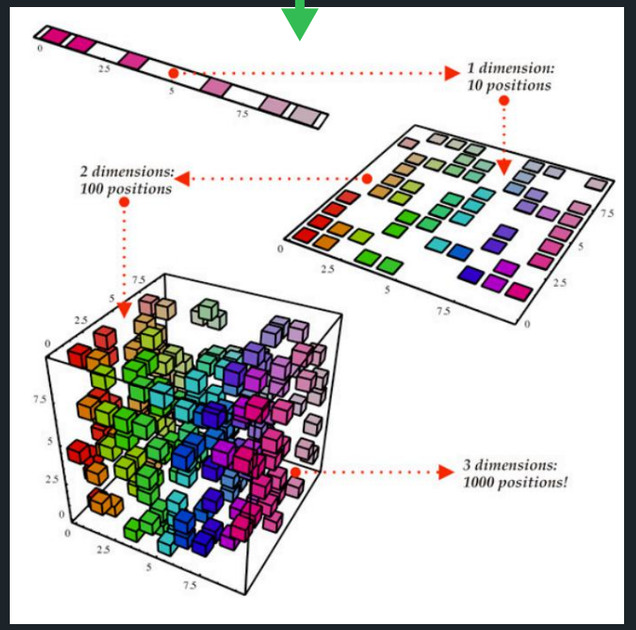


Investigate all relevant features



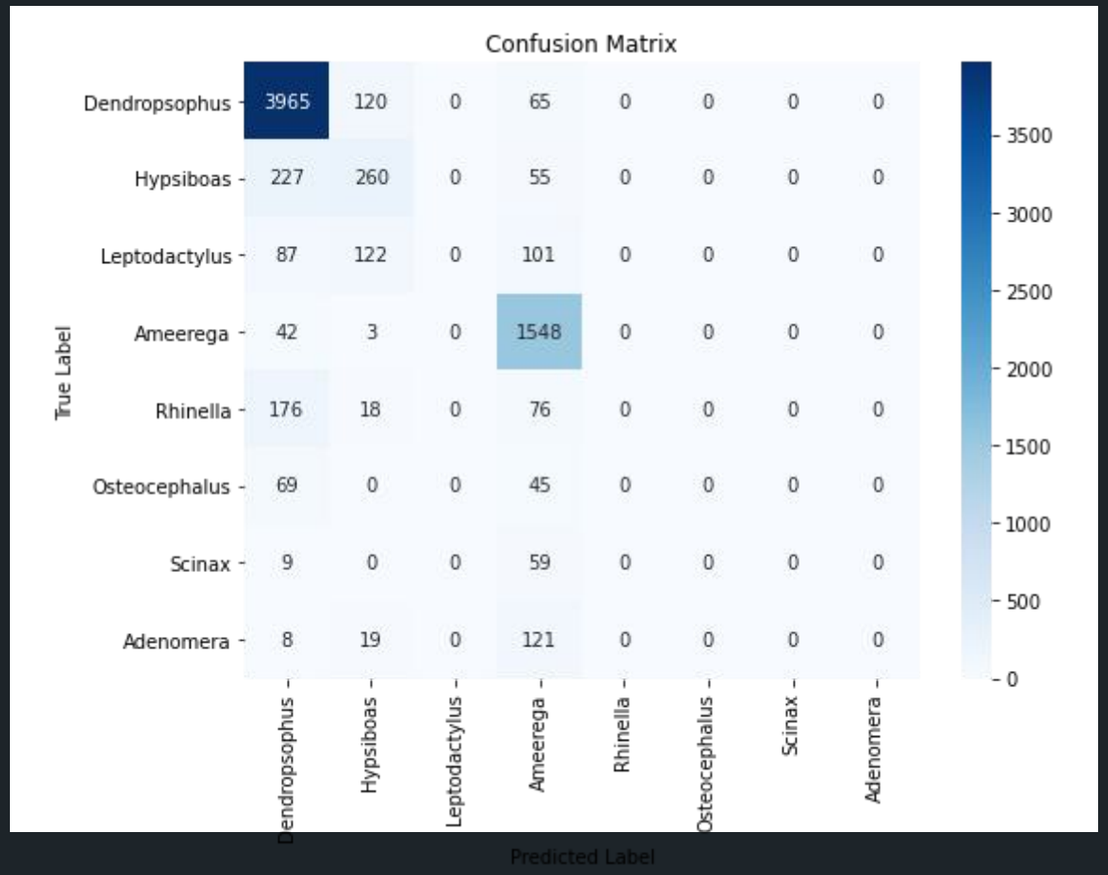
WE LEARN TO FOLLOW CERTAIN STEPS PRIOR TO CLUSTERING

load data → clean data → standardise data → reduce dimensions? → fit cluster algorithm



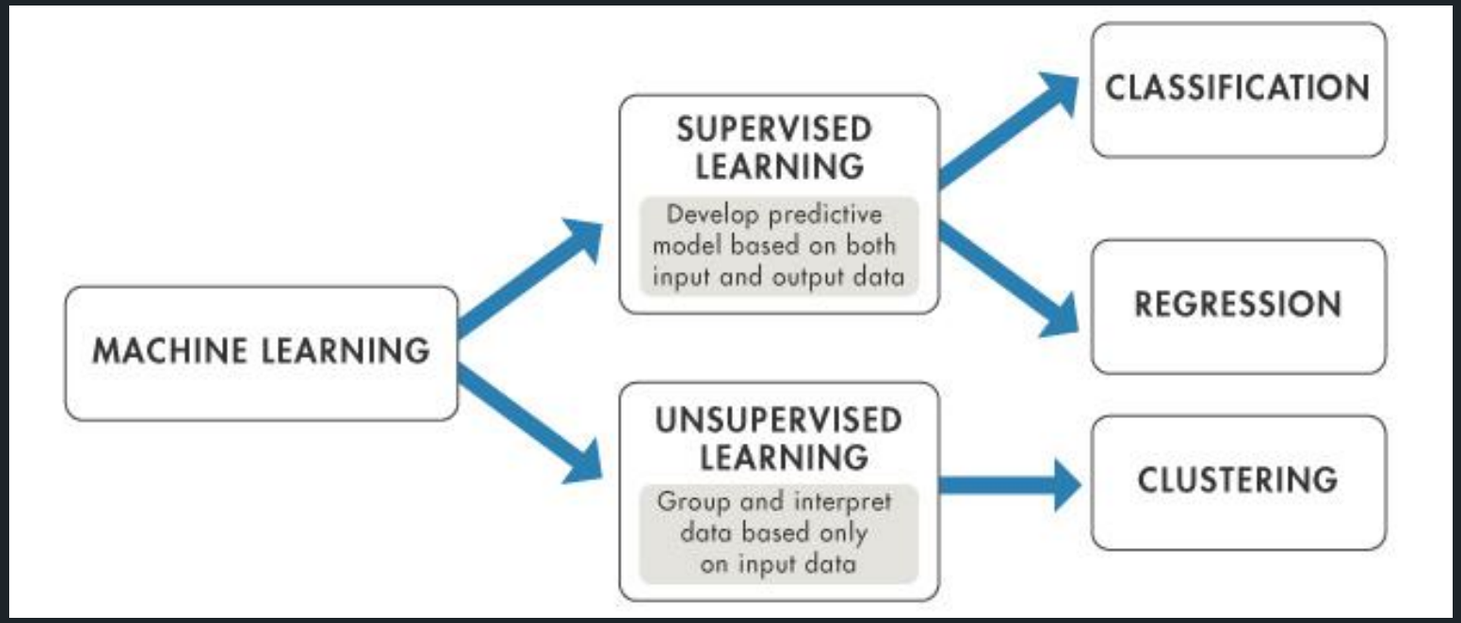
WE LEARN TO FOLLOW CERTAIN STEPS PRIOR TO CLUSTERING...[GREAT IF WE HAVE KNOWN ANSWERS]

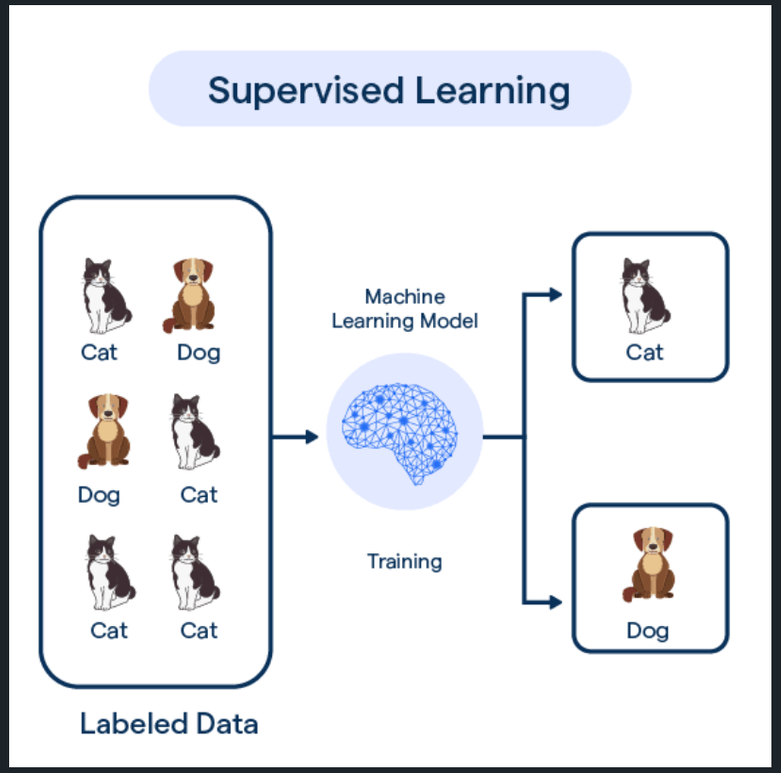
load data → clean data → standardise data → reduce dimensions? → fit cluster algorithm → evaluate accuracy





SUPERVISED LEARNING....

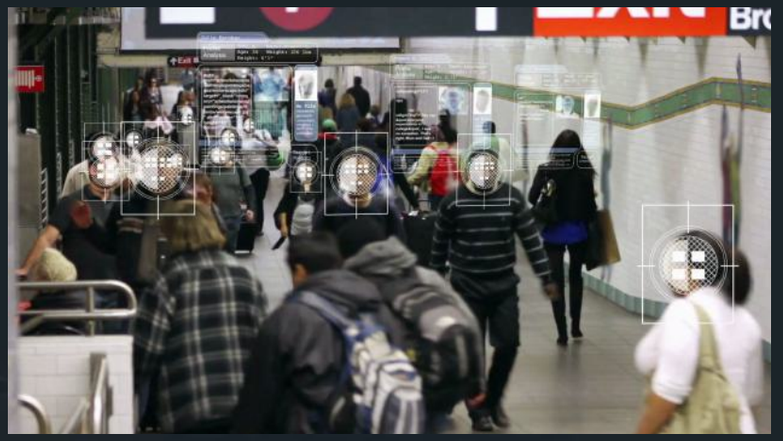




PANDAS AND DATA DISCOVERY



- MANIPULATING DATAFRAMES AND STATISTICAL ANALYSIS
- PRACTICAL 8



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